

Energy Storage System (ESS)

Team MDS8

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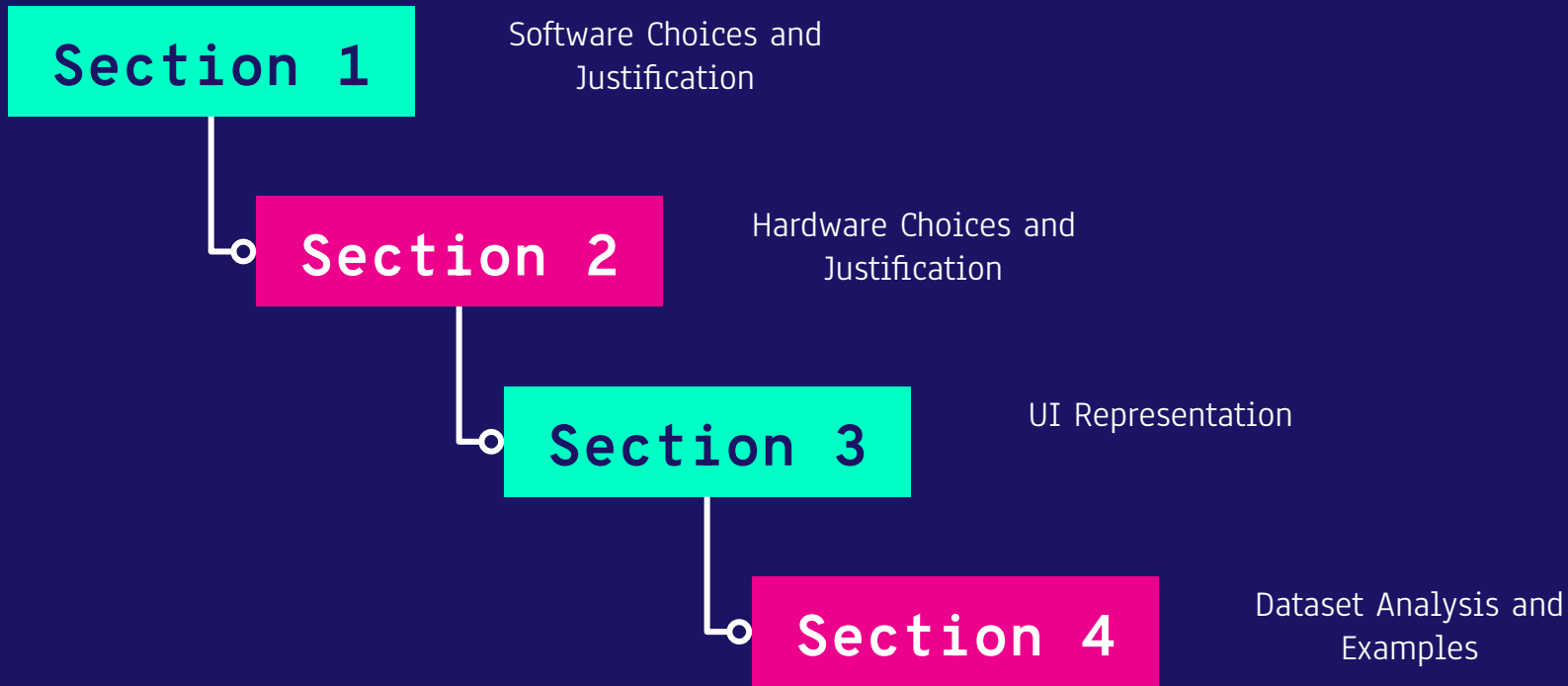
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Introduction

- The global pursuit of sustainable and efficient energy management solutions has led to the exploration of advanced technologies that can revolutionise the energy sector.
- In this context, the integration of artificial intelligence (AI) techniques holds significant promise for optimising energy utilisation, enhancing grid stability, and maximising the return on investment (ROI) of energy storage systems (ESS).

Presentation Flow

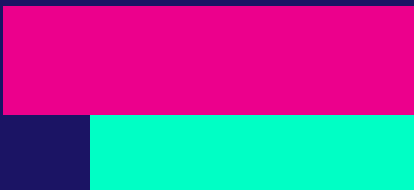




Software Choices and Justification



Section 1



Operating System

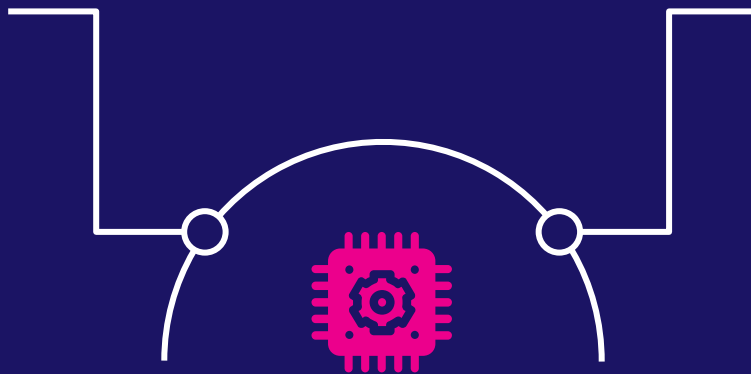
Linux

Linux for server-based
deployments.

Ubuntu,CentOS

Windows

Windows for development.



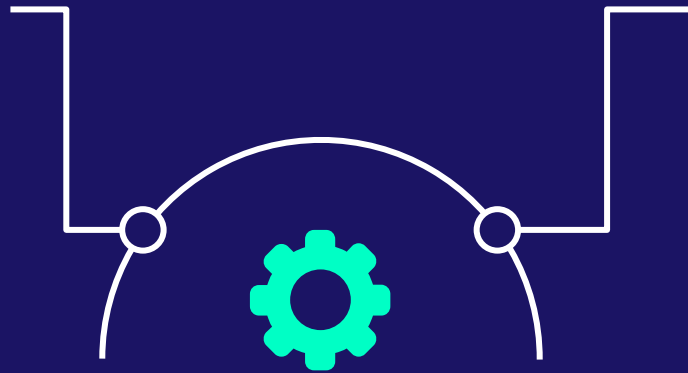
Database Management System

**Data storage
and
retrieval**

PostgreSQL, MySQL, or
SQLite

**NoSQL
databases**

MongoDB



Web Development

**Server
software**



Apache
Nginx

**Backend
framework**



Flask for Python
Express.js for Node.js

**Frontend
framework**



React
Angular

Machine Learning and Data Analysis

Machine
learning

Scikit-learn
TensorFlow



Jupyter Notebook

IDE

Data Processing

data preprocessing
and manipulation



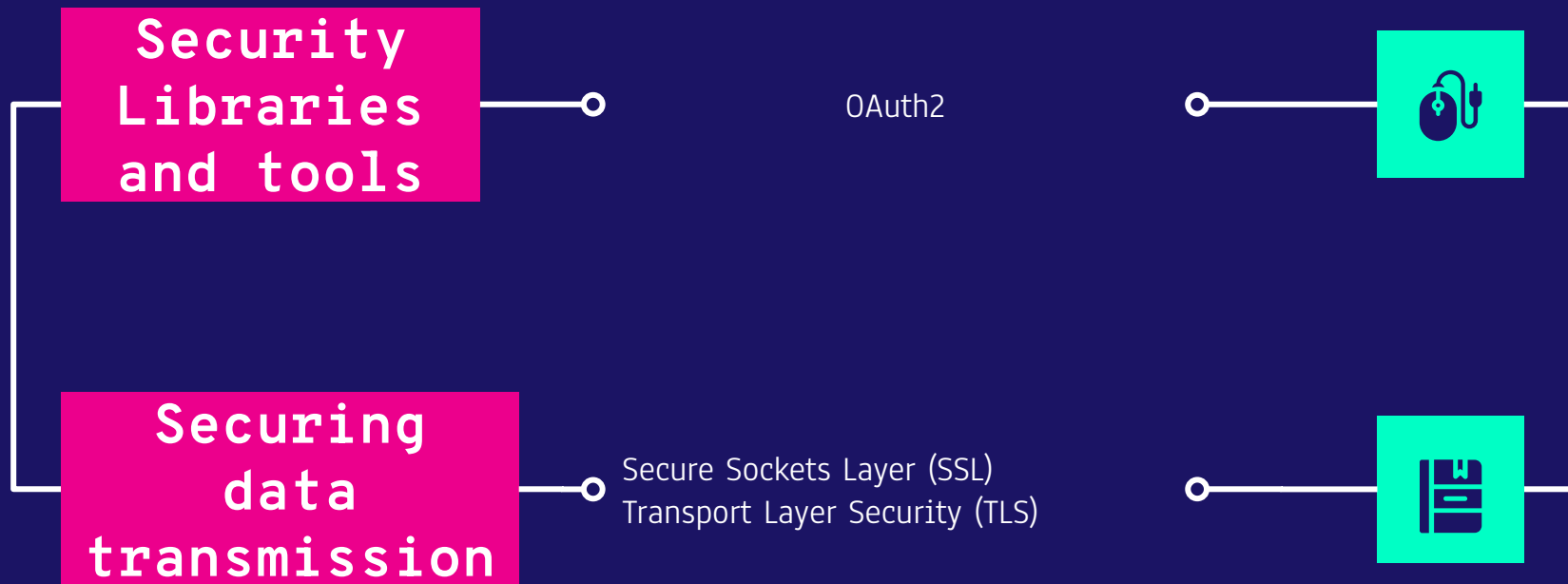
Pandas

Time-series analysis
libraries



Statsmodel

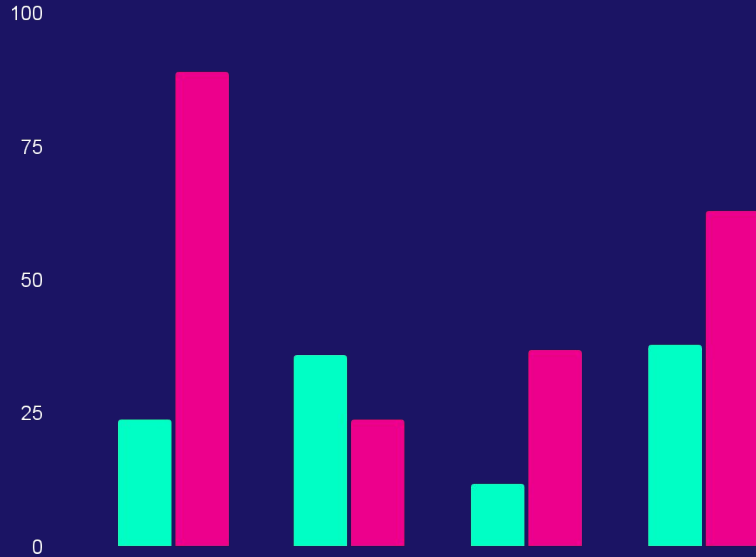
Security



Containerization and Deployment



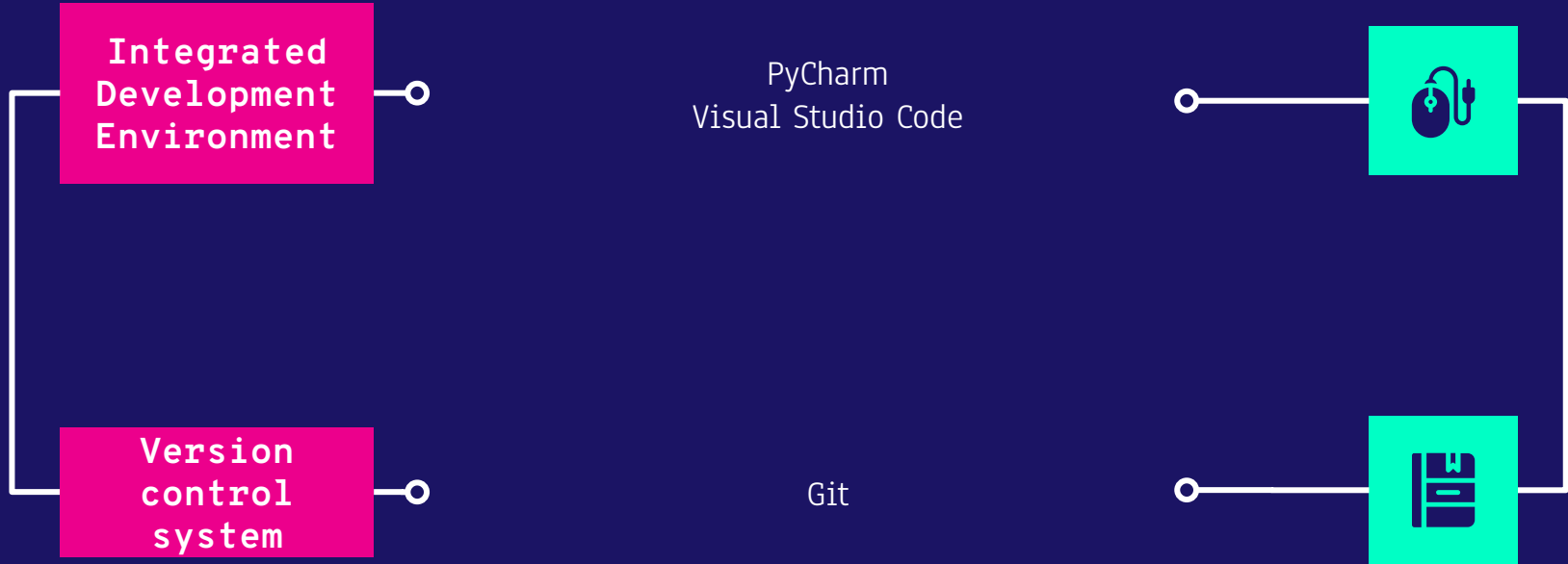
Visualization



**Data
Visualisation**

Matplotlib

Development Tools

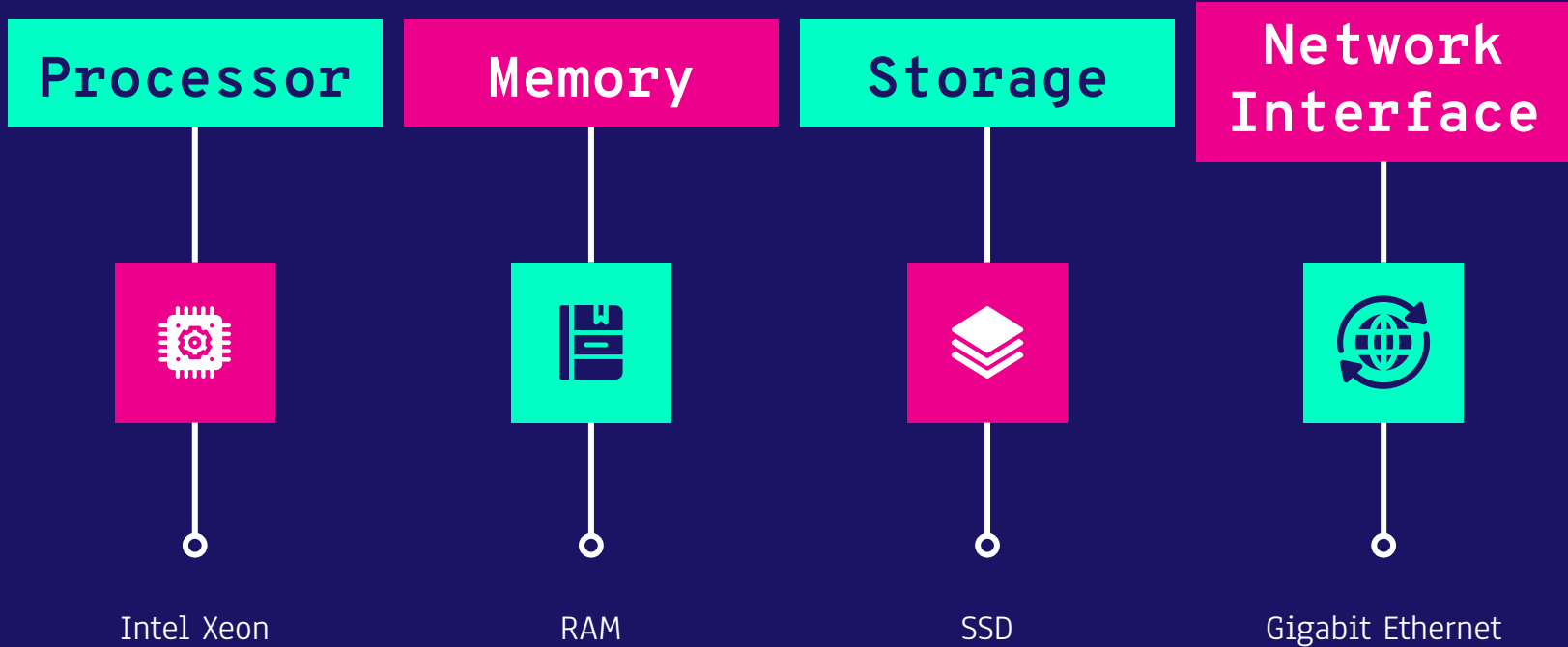




Hardware Choices and Justification

Section 2

Server Hardware



For Client Devices, a standard desktop, laptop, or mobile device that can access the internet would be sufficient.

Data Collection Hardware

Sensor



smart metres
data loggers

Cloud Services

AWS

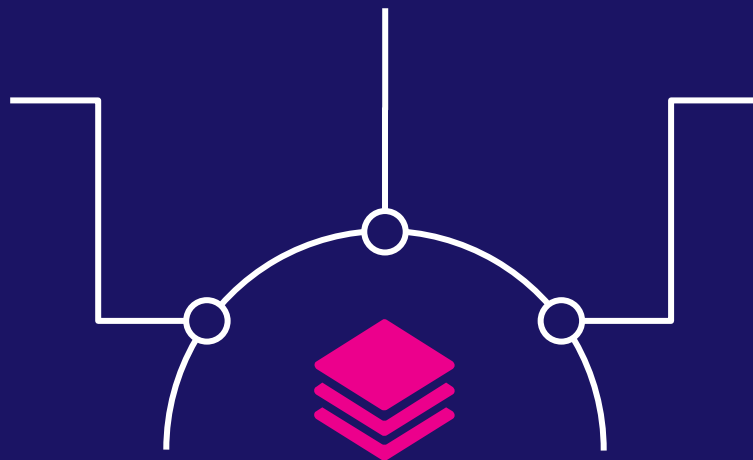
(Amazon Web Services)

Azure

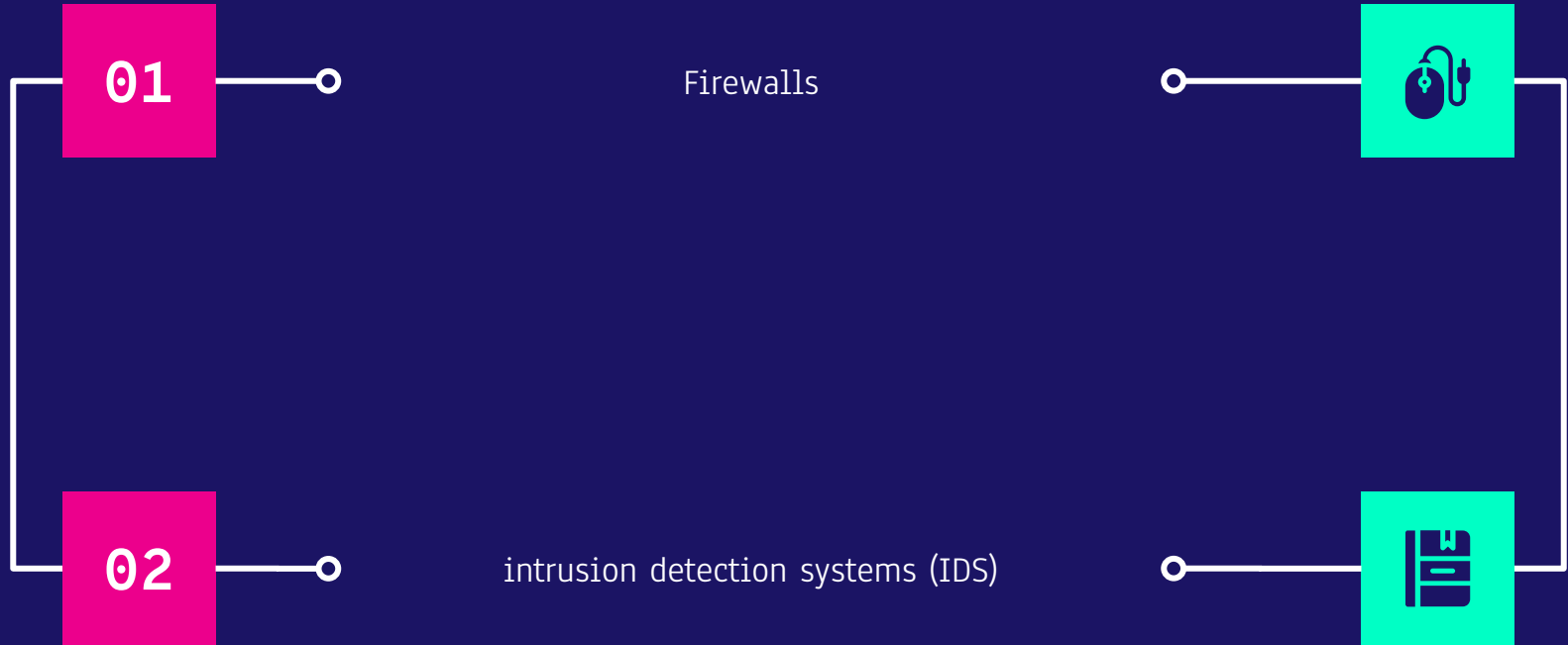
(Microsoft Azure)

GCP

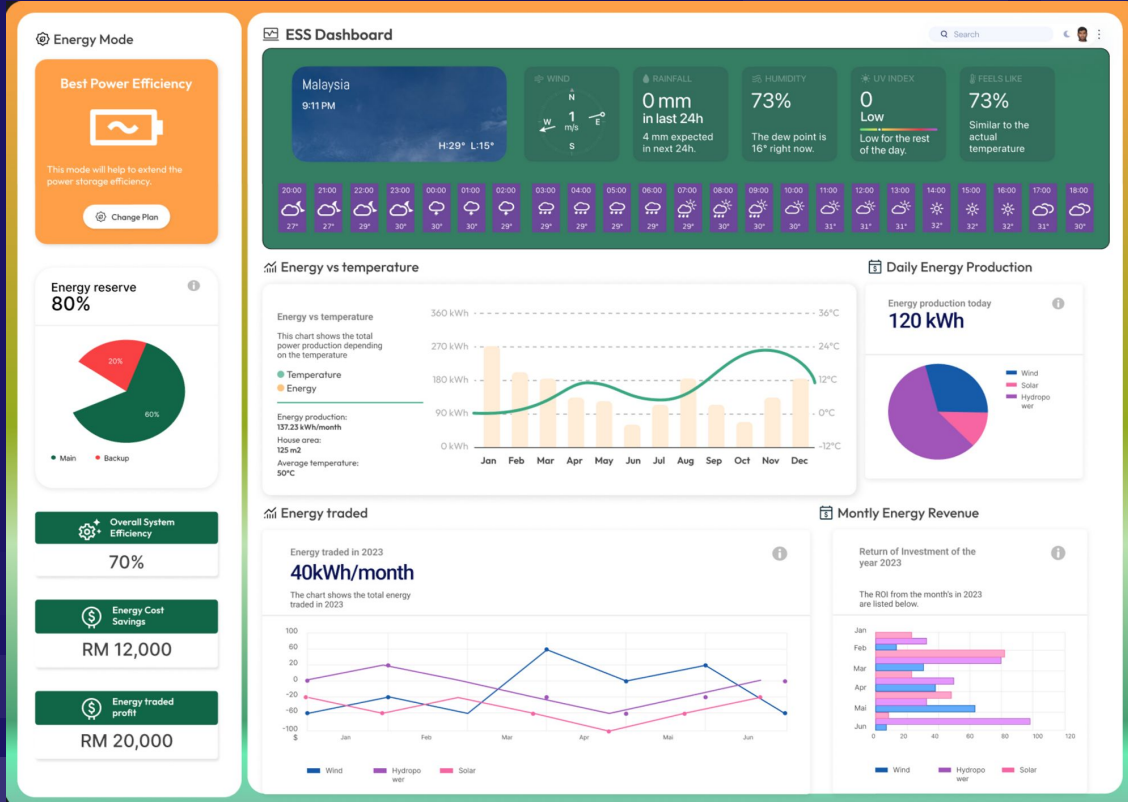
(Google Cloud Platform)



Security Hardware



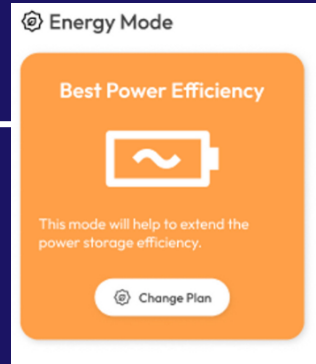
UI Representation



UI Representation

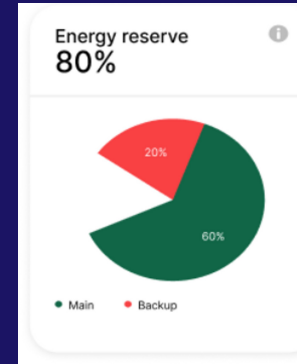
Energy Mode

Current selected mode by the client.



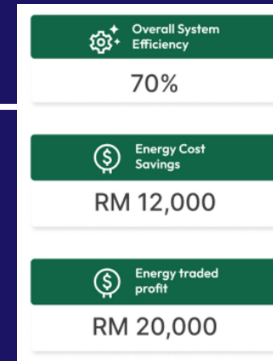
Energy Reserve

the stored energy that is available for consumption or trading.



System Performance

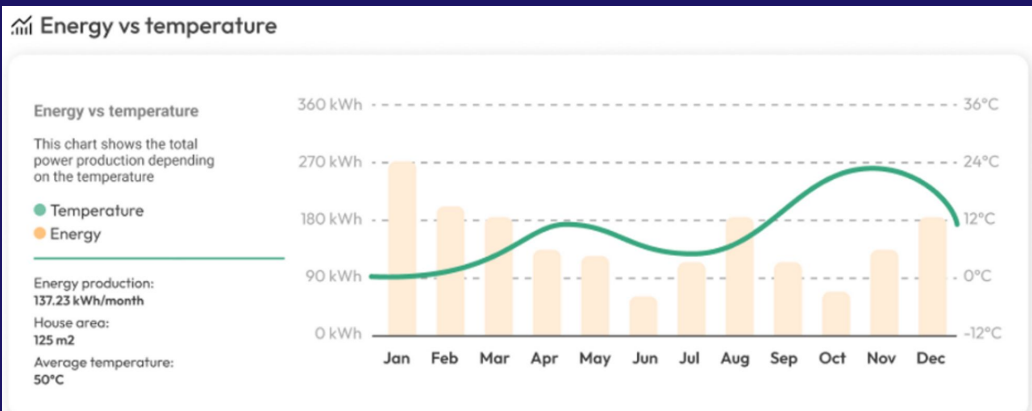
Represent few statistics of how the system performs.



UI Representation

Weather information

beneficial that the client view weather information.



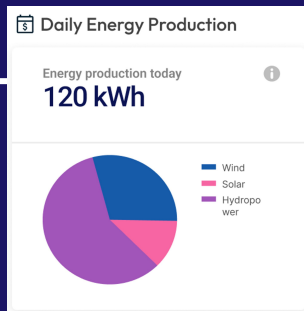
Energy vs temperature chart

demonstrates how temperature influences on energy production

UI Representation

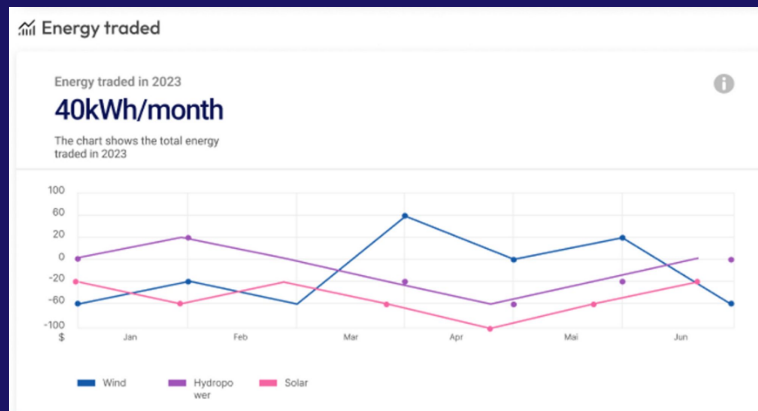
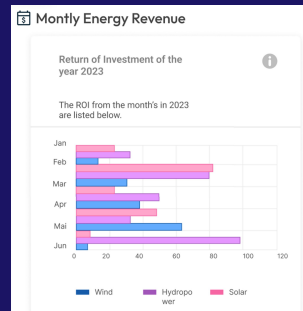
Daily production

Daily production categorised by mode of production



Monthly Energy revenue

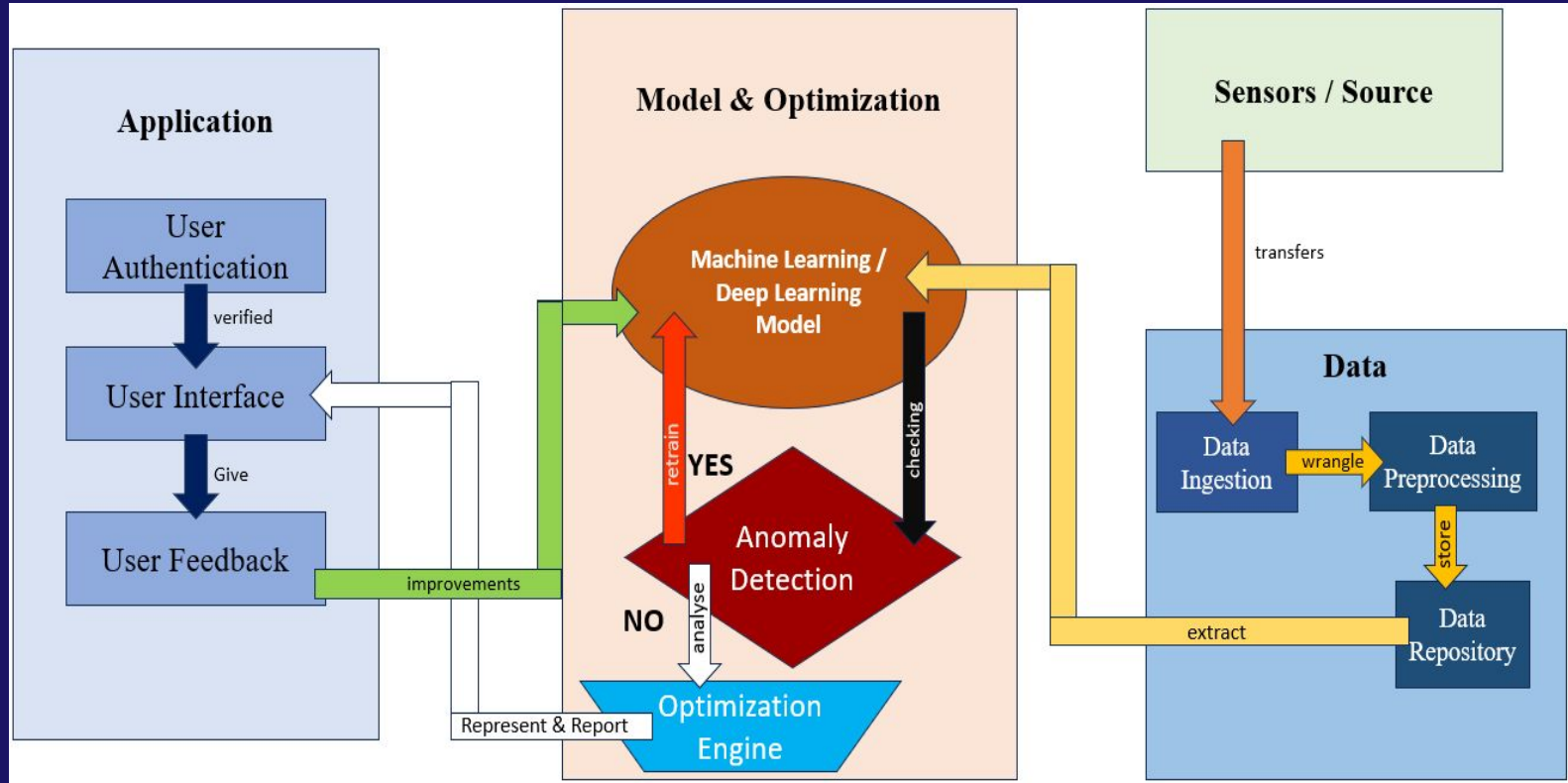
Client's profit based on each month and the energy production method

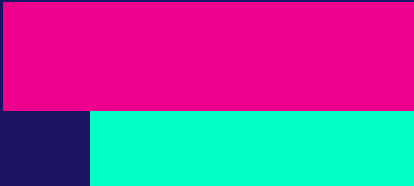
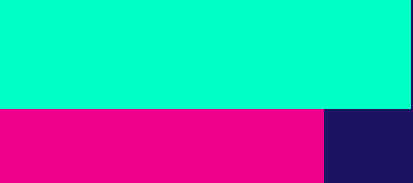


Energy Traded

Amount of energy traded and that will be traded.

System Architecture





Data Analysis

Section 3



Data Analysis Summary

Objective	Perform data cleaning, exploration, and analysis, conduct inferential statistics, and present findings to prepare the dataset for AI load balancing.	
Methodology	Utilize RStudio and Kaggle for data work, benefiting from Kaggle's diverse datasets, community support, data quality, and real-world relevance	
Data Preprocessing	Steps include removing invalid values, converting data types, and transforming data type to ensure data consistency and readiness for analysis.	
Data Quality Check	Verify data consistency, address missing values, remove invalid rows, and ensure accurate data types for data integrity.	

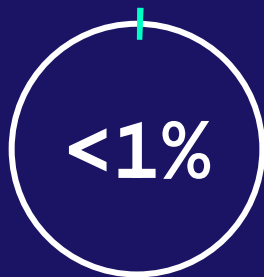
Sample Data Preview

	Date	Voltage	Global Active Power	Sub-metering 1
1	2021-01-01 00:00:00	255	255.06	0.14
2	2021-01-01 00:00:01	260	266.07	0.12
3	2021-01-01 00:00:02	280	268.14	0.34
4	2021-01-01 00:00:03	260	290.20	0.26

P-value and 95% confidence interval

P-value

6.486411e-11

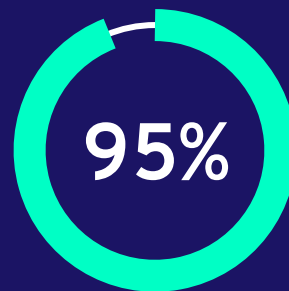
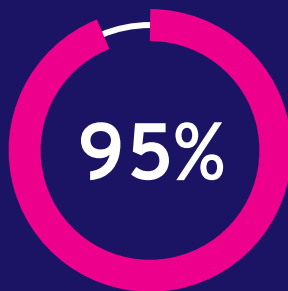


Afternoon

(1.4507515, 1.52590747)

Evening

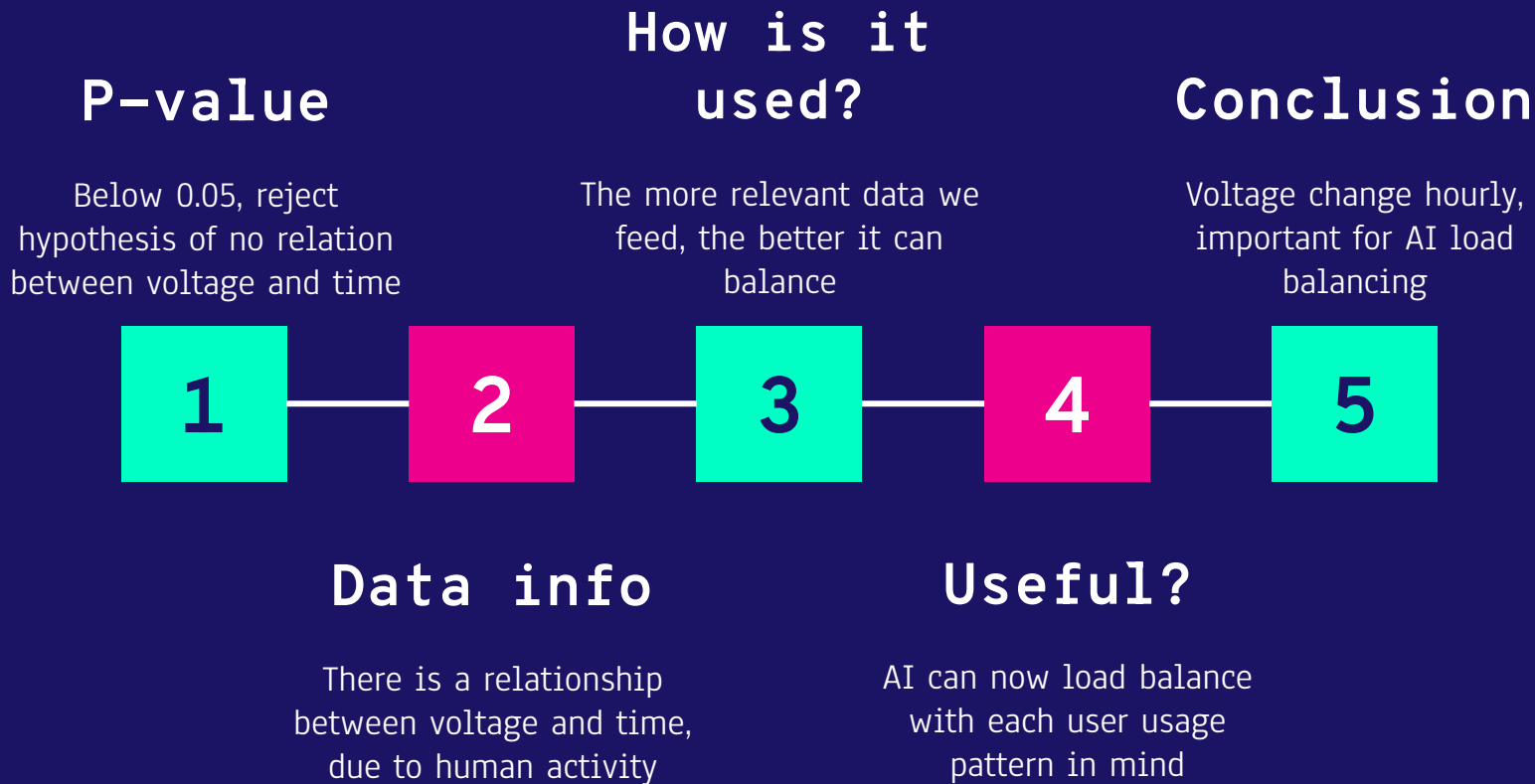
(-0.1627477, -0.08762552)



Night

(2.3620281, 2.43723847)

What does it imply?





Thank you for listening!



Q & A session

